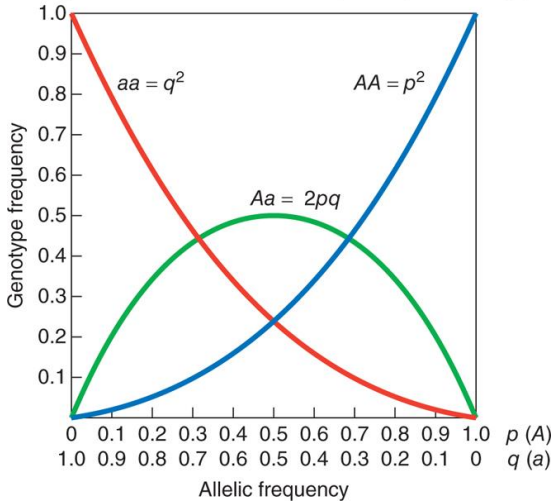
BME 105: Genetics in the Genomics Era

Lecture 20: Intro to Population Genetics



Instructor: Dr. Shelbi Russell,
Assistant Professor

Course website on canvas.ucsc.edu

Lecture 20 Learning Objectives

- Be able to define the population genetics terms: population, gene pool, sample
- Be able to distinguish genotype frequencies from phenotype frequencies, and calculate one from the other
- Know and be able to list the five assumptions required of populations at Hardy-Weinberg Equilibrium (HWE): random mating, no mutations, no gene flow, no natural selection, infinite population size (no drift)
- Understand why a population in HWE is <u>not</u> evolving
- Be able to calculate HWE genotype frequencies: $1 = p^2 + 2pq + q^2$ (i.e., the binomial expansion of biallelic possibilities)
- Be able to perform a chi-squared test to determine whether genotypes from nature are in HWE, or not.
- Be able to articulate what genetic drift is and why the size of its effects are linked to population size
- Define allele fixation
- What is a founder effect in population genetics?

Terms used to describe populations

Population– a group of interbreeding individuals of a single species living in the same time and place

Gene pool– the total of all alleles carried in all members of a population

Sample – a number of individuals used to make inferences about the entire population

Phenotype vs genotype frequency

Phenotype frequency – proportion of individuals in a population that exhibit a particular phenotype

E.g., if a is recessive to A, then the frequency of the aa phenotype is ¼ = genotype frequency

Genotype frequency – proportion of individuals in a population that have a particular genotype

E.g., frequency of aa = aa / (aa + Aa + aA + AA)

Allele frequency - proportion of each allele across the total haplotypes in a population

E.g., frequency of a = a / (a+A)

⇒ Allele frequencies can be obtained from genotype frequencies

Calculating allele frequencies from genotype frequencies

Phenotype frequency – proportion of individuals in a population that exhibit a particular phenotype

Genotype frequency – proportion of individuals in a population that have a particular genotype

Allele frequency - proportion of each allele across the total haplotypes in a population

Example: A gene with two alleles (A and a) in a population of 20 individuals

12 individuals are AA → 24 A alleles

4 individuals are Aa → 4 A alleles and 4 a alleles

4 individuals are aa → 8 a alleles

Genotype	*AA*	*Aa*	*aa*	Total
Number of individuals	12	4	4	20
Genotype frequency	0.6	0.2	0.2	1.0

Allele	*A*	*a*	Total
Number of chromosomes	28	12	40
Allele frequency	0.7	0.3	1.0

Calculate <u>allele</u> frequencies:

A alleles = (24 + 4)/40 = 0.7

a alleles = (8 + 4)/40 = 0.3

General formula for calculating allele frequency

Frequency of A =

frequency of AA + (1/2) frequency of Aa

Frequency of a =

frequency of aa + (1/2) frequency of Aa

Genotype	*AA*	*Aa*	*aa*	Total
Number of individuals	12	4	4	20
Genotype frequency	0.6	0.2	0.2	1.0

Allele	*A*	*a*	Total
Number of chromosomes	28	12	40
Allele frequency	0.7	0.3	1.0

Hardy-Weinberg Equilibrium (HWE) - genetic variation will remain constant across generations in an ideal population

Developed independently in 1908 by G.H. Hardy and W. Weinberg

Five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- Infinitely large population (no drift)

⇒ If assumptions are met, the population is <u>not evolving</u>

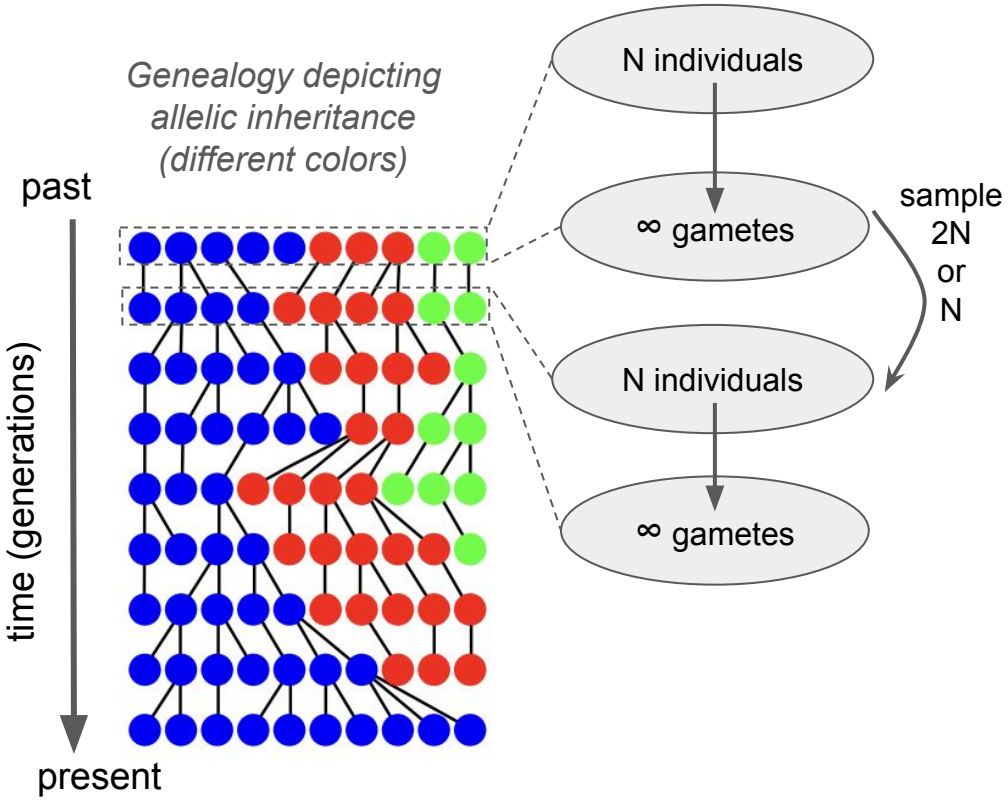
Hardy-Weinberg Equilibrium (HWE) - genetic variation will remain constant across generations in an ideal population

Five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- Infinitely large population (no drift)

***If nothing influences allele frequencies, then they will be equal from one generation to the next*

⇒ genotypes will be based on allele frequencies + random assortment



Deriving Hardy-Weinberg Equilibrium (HWE) allele and genotype frequencies

Five simplifying assumptions:

- Random mating
- No mutation
- No gene flow (no migration)
- No selection (no difference in survival or fitness)
- Infinitely large population (no drift)

Population allele frequencies can be estimated

For a biallelic locus:

2 alleles at a locus/site

A a

2 allele frequencies at a locus/site

p q

3 possible genotypes in a **diploid**:

AA Aa aa

3 genotype frequencies in a **diploid**:

x_{AA} x_{Aa} x_{aa}

Table 1: Punnett square for Hardy–Weinberg

		Females	
		A (*p*)	**a (*q*)**
Males	**A (*p*)**	AA (p^2)	Aa (pq)
	a (*q*)	Aa (qp)	aa (q^2)

Genotype frequencies: $x_{AA} + x_{Aa} + x_{aa} = 1$

Sample allele frequencies:
A frequency → $p = x_{AA} + ½\, x_{Aa}$
a frequency → $q = 1 - p = x_{aa} + ½\, x_{Aa}$

HWE genotype frequencies: $p^2 + 2pq + q^2 = 1$

(Expanded HWE genotypes: $pp + pq + qp + qq = p^2 + 2pq + q^2 = 1$)

Predicting genotype frequencies from allele frequencies with HWE (another example)

Assume a large population of randomly breeding individuals with no new mutations, no migration, and no differences in fitness based on genotype.

If in HWE, the population's genotype frequencies will sum up to 1 using this formula:

$$p^2 + 2pq + q^2 = 1$$

p = frequency (A)

q = frequency (a)

Allele frequencies
p = freq. (A) = 0.7
q = freq. (a) = 0.3

Sperm \ Eggs	p 0.7	q 0.3
p 0.7	AA 49%	Aa 21%
q 0.3	Aa 21%	aa 9%

Homozygote $AA = \boldsymbol{p^2} = 0.49$
Heterozygote $Aa = \mathbf{2(pq)} = 0.42$
Homozygote $aa = \boldsymbol{q^2} = 0.09$
1.00

Departure from HWE serves as evidence for evolution

1. Measure genotypes in nature
2. Calculate allele frequencies from genotypes

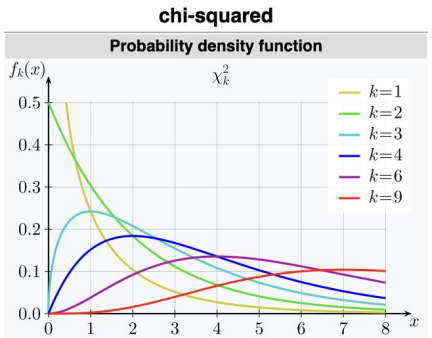
   A_1 frequency → $p = x_{11} + ½\, x_{12}$
   A_2 frequency → $q = 1 - p = x_{22} + ½\, x_{12}$

3. Calculate genotype frequencies expected under HWE
4. Compare genotypes under the **null distribution (HWE) <u>to</u> nature**

   e.g., with the chi-squared test

The chi-squared distribution (with k degrees of freedom) = $Q = \sum_{i=1}^{k} Z_i^2,$

distribution of a sum of the squares of k independent standard normal random variables.



Departure from HWE serves as evidence for evolution

1. Measure
2. Calculate

   A_1 frequency
   A_2 frequency

3. Calculate
4. Compare

   e.g., w

Example chi-squared test for deviation [edit]

This data is from E. B. Ford (1971) on the scarlet tiger moth, for which the phenotypes of a sample of the population were recorded. Genotype-phenotype distinction is assumed to be negligibly small. The null hypothesis is that the population is in Hardy–Weinberg proportions, and the alternative hypothesis is that the population is not in Hardy–Weinberg proportions.

Table 3: Example Hardy–Weinberg principle calculation

Phenotype	White-spotted (AA)	Intermediate (Aa)	Little spotting (aa)	Total
Number	1469	138	5	1612

From this, allele frequencies can be calculated:

$$p = \frac{2 \times \text{obs}(\text{AA}) + \text{obs}(\text{Aa})}{2 \times (\text{obs}(\text{AA}) + \text{obs}(\text{Aa}) + \text{obs}(\text{aa}))}$$

$$= \frac{2 \times 1469 + 138}{2 \times (1469 + 138 + 5)}$$

$$= \frac{3076}{3224}$$

$$= 0.954$$

and

$$q = 1 - p$$
$$= 1 - 0.954$$
$$= 0.046$$

So the Hardy–Weinberg expectation is:

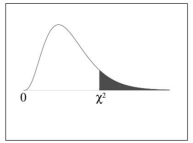
$$\text{Exp}(\text{AA}) = p^2 n = 0.954^2 \times 1612 = 1467.4$$
$$\text{Exp}(\text{Aa}) = 2pqn = 2 \times 0.954 \times 0.046 \times 1612 = 141.2$$
$$\text{Exp}(\text{aa}) = q^2 n = 0.046^2 \times 1612 = 3.4$$

Pearson's chi-squared test states:

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
$$= \frac{(1469 - 1467.4)^2}{1467.4} + \frac{(138 - 141.2)^2}{141.2} + \frac{(5 - 3.4)^2}{3.4}$$
$$= 0.001 + 0.073 + 0.756$$
$$= 0.83$$

df = 1 for two alleles, one locus

Chi-Square Distribution Table



The shaded area is equal to α for $\chi^2 = \chi^2_\alpha$.

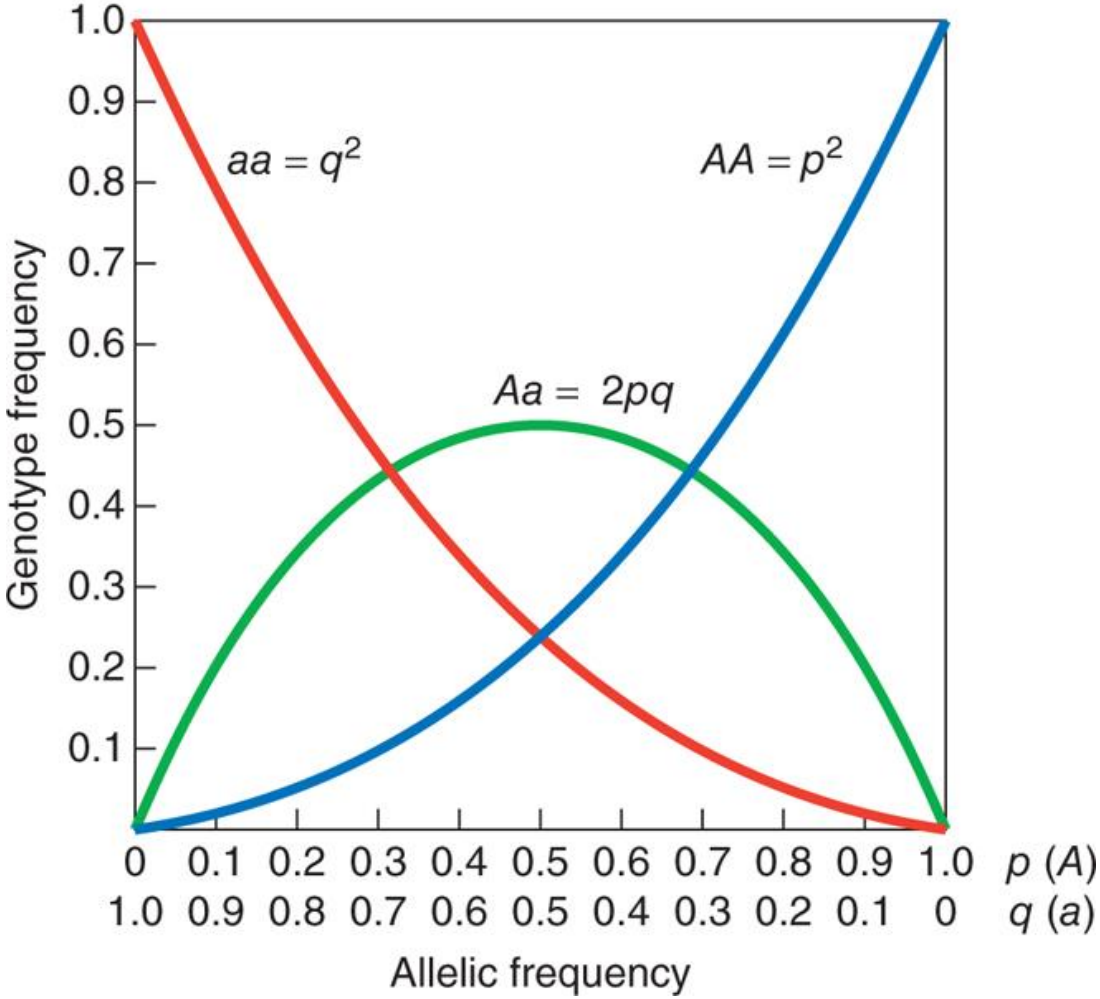
df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750

p-value between 0.1 and 0.9 ⇒ *not significantly different from HWE expectations*

For any set of allele frequencies, only one set of genotype frequencies results in HWE

Allele frequencies must sum to 1

Assumptions of HWE have to hold (only for alleles in HWE)



How do you get changes in allele frequencies?

HWE is accurate in predicting allele frequency in the short term (generation-to-generation) rather than the long term (macroevolution)

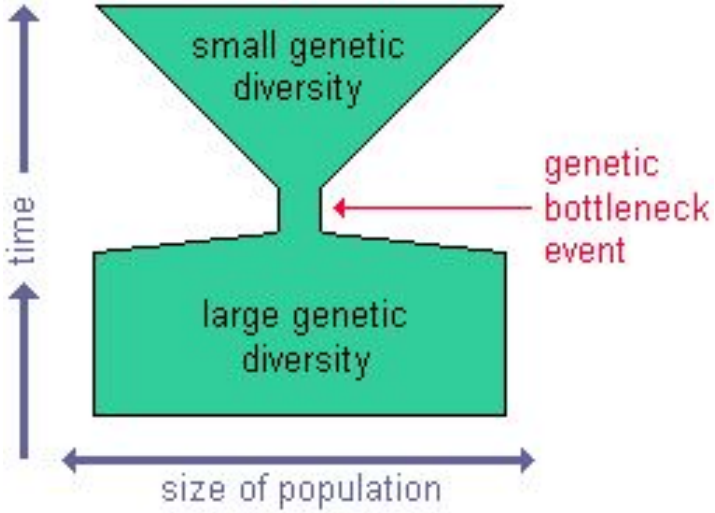
Violations of the 5-assumptions will cause changes in allele frequency:

- **Finite population sizes (genetic drift)**
    - Temporarily small population sizes
    - Fragmented (meta) populations
    - Long-term small populations

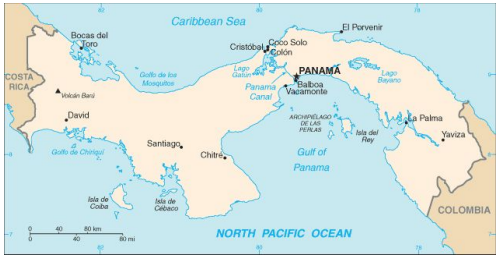
Genetic drift: an issue if population size isn't ~infinite

<u>Bottlenecks (temporary reductions in population size)</u>

- Local near-extinction events
- Founder effects

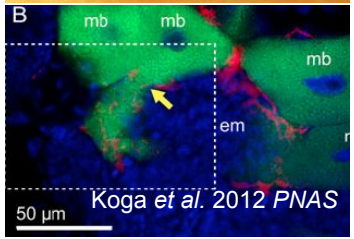


- Intracellular symbiont inheritance
- Vicariance events (*e.g.*, the formation of the isthmus of Panama)



Genetic drift: a permanent issue if small population size doesn't recover

<u>Small populations (permanent)</u>



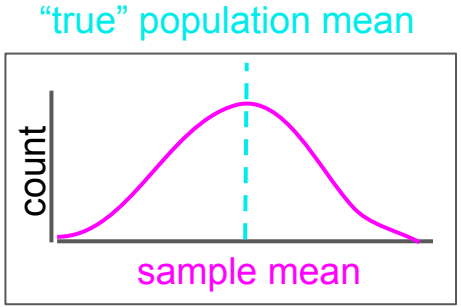
Koga *et al.* 2012 *PNAS*

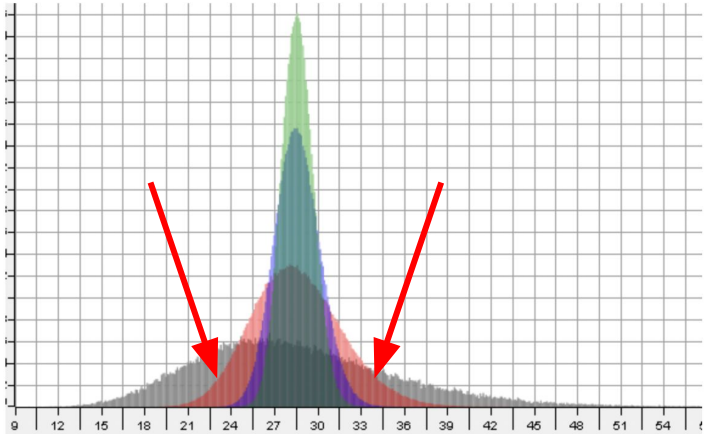
Bailey 2007 *Heredity*

Genetic drift: a living manifestation of high variance in small sample sizes

<u>Random chance has an outsized effect when N is small</u>

→ related to the central limit theorem





In the graph above, the gray color shows the skewed distribution of the values in the population. The other colors represent the sampling distributions of the means for different sample sizes. The red color shows the distribution of means when your sample size is 5. Blue denotes a sample size of 20. Green is 40. The red curve (n=5) is still skewed a bit, but the blue and green (20 and 40) are not visibly skewed.

https://statisticsbyjim.com/basics/central-limit-theorem/

small populations are like small sample sizes:

→ higher variance "draws" each generation

⇒ wider sampling error

Genetic drift: low-probability events strongly influence small populations

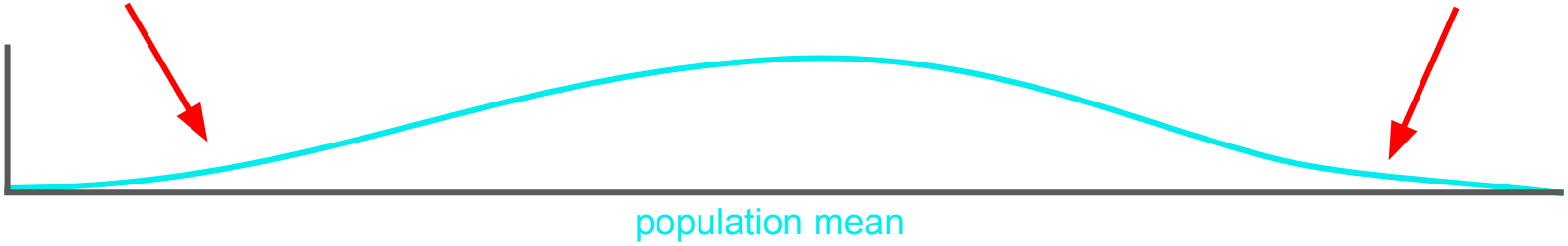
<u>Random chance has an outsized effect when N is small</u>

If p = q = ½ in a population in HWE, genotype frequencies will be ¼ AA, ½ Aa, ¼ aa

Example scenario, a bottleneck that leaves only one allele permanently purges the other:

A bottleneck that leaves only 4 AA has a probability of $¼^4$ = 1/256 = <u>3.9e-3</u>

A bottleneck that leaves 40 AA has a probability of $¼^{40}$ = 1/1.21e24 = <u>8.3e-25</u>



Monte Carlo simulations to model long-term changes in allele frequencies with finite population size

Monte Carlo simulations use a computer program to model possible outcomes of randomly chosen matings over a designated number of generations

- Starting population has a defined number of individuals that are homozygous and heterozygous
- Mating pairs are chosen through a random-number generating program
- Genotypes of offspring at each generation are based on probabilities

Monte Carlo simulations to model long-term changes in allele frequencies with finite population size

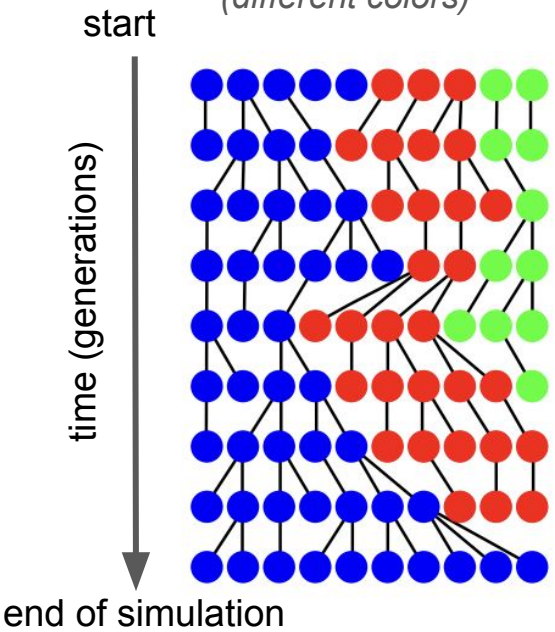
At each generation in the simulation:

- Total offspring number and parental population size are equal
- Parental generation is discarded and offspring serve as parents of next generation

Multiple, independent simulations are performed

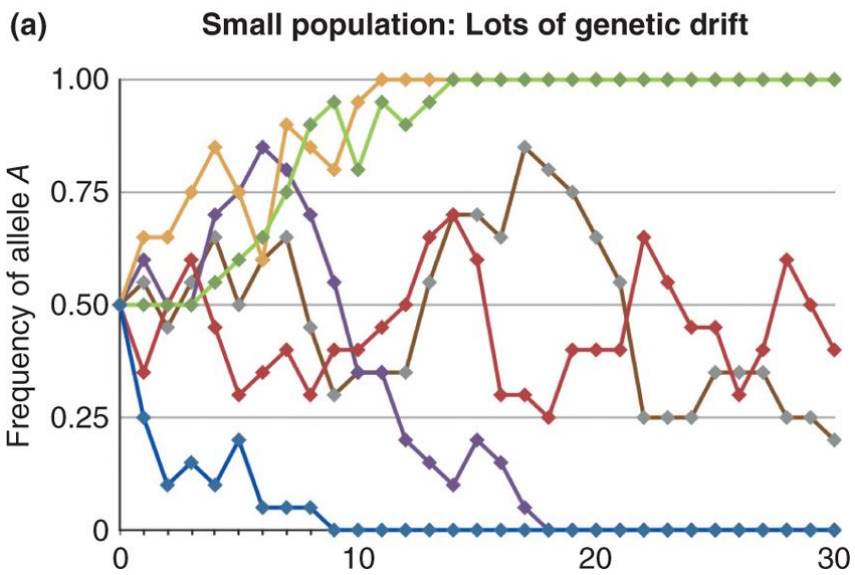
Each simulation represents a possible pathway of **genetic drift** (changes in allele frequency due to random chance, not due to selection)

Genealogy depicting allelic inheritance (different colors)



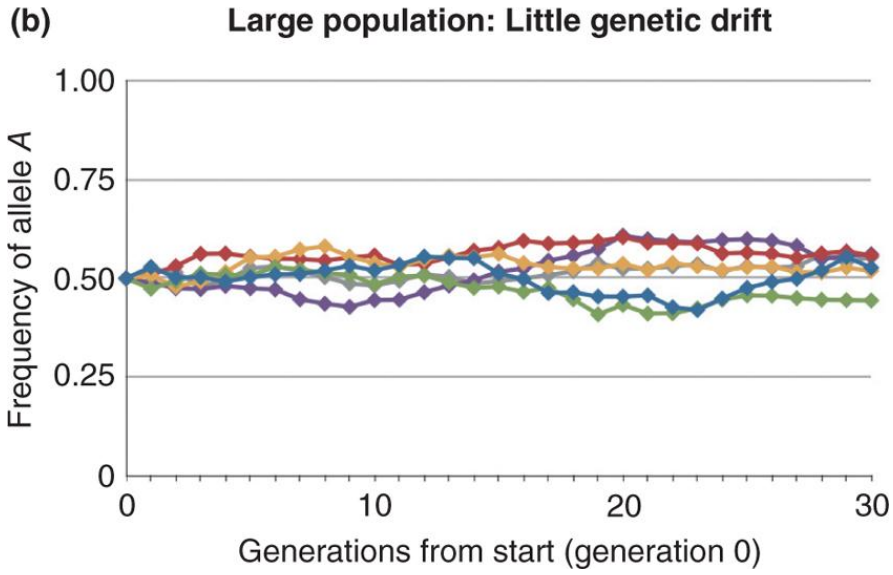
Monte Carlo modeling of population drift in populations of different sizes

Initial population has 10 individuals



Each color line is a different result from a simulation

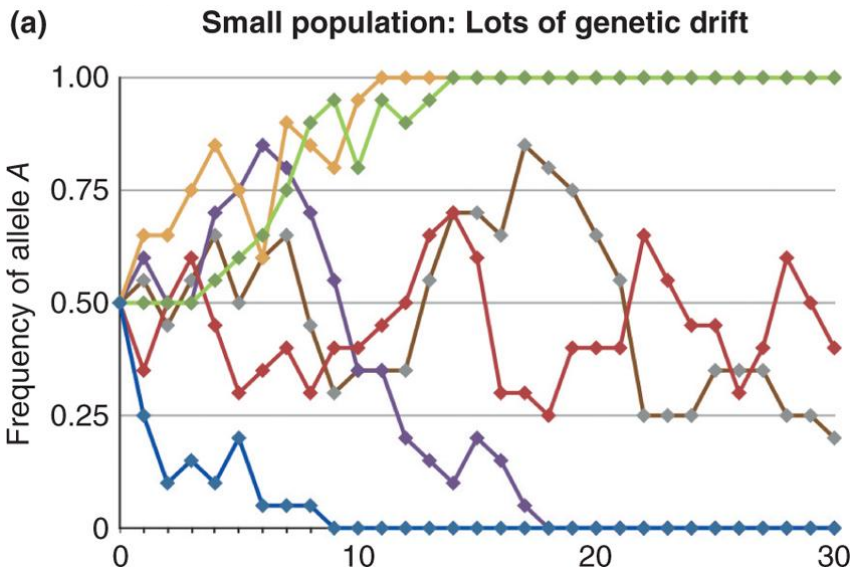
Initial population has 500 individuals



HWE assumptions met ⇒ allele frequencies don't change

Monte Carlo modeling of population drift in populations of different sizes

Initial population has 10 individuals



Fixation - when only one allele in a population has survived

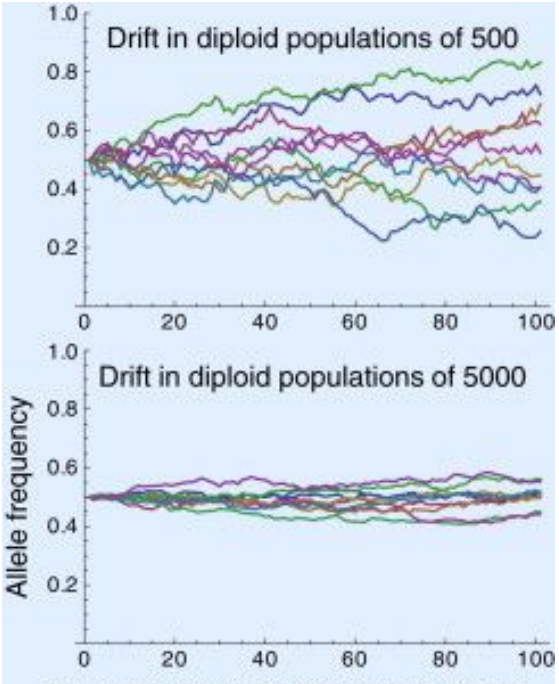
Loss - when allele is removed from the population

Genetic drift manifests in large allele frequency changes from generation-to-generation

Drift

$\propto 1/N$

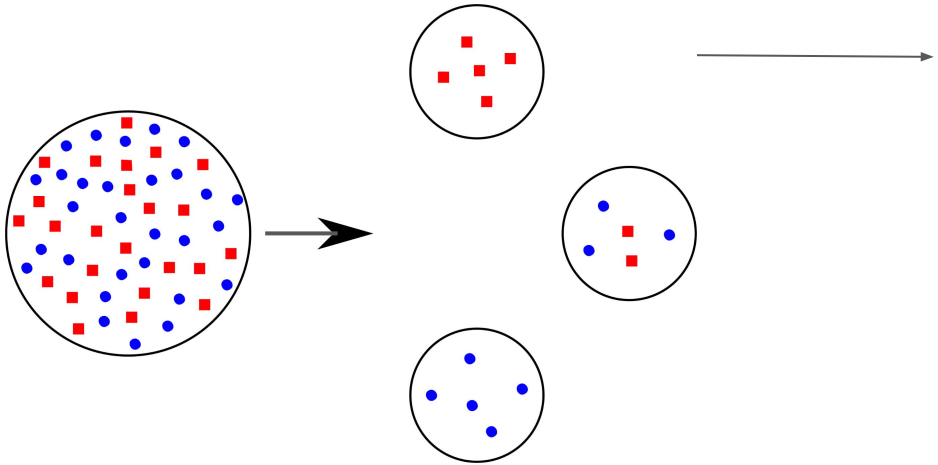
What is genetic drift? Say you have a population of 5,000 people. That makes 10,000 copies of each gene. Imagine a gene where 3,000 of those copies are of one particular allele or type. In the next generation, there won't necessarily be exactly 3,000 copies again. There may be 3,050 or 2,960 copies instead. Some gametes get randomly picked out of all the possible gametes that could have been used. This is a bit like tossing a coin, and 100 coin tosses rarely yield exactly 50 heads. Natural selection happens when individuals developed from certain gametes are more likely to survive and reproduce. Genetic drift, together with mutation and recombination, randomly produces the gametes that selection can act on. Or, if there is no selection, allele frequencies can change by mutation and genetic drift alone.



Masel 2011 *Current Biology*

Genetic drift can cause a higher prevalence of a disease in isolated populations - “founder effect”

Old Order Amish, of Lancaster County, Pennsylvania



Larger population

Smaller isolated populations

High prevalence of Ellis-van Creveld syndrome which is caused by an autosomal recessive mutation. Rare in broader population.

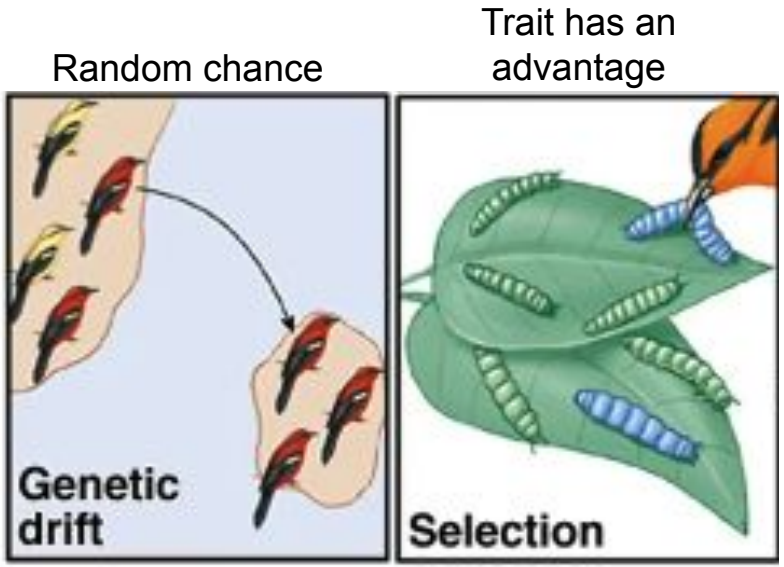
How do you get changes in allele frequencies?

HWE is accurate in predicting allele frequency in the short term (generation-to-generation) rather than the long term (macroevolution)

Violations of the 5-assumptions will cause changes in allele frequency:

- Finite population sizes (genetic drift)
- **<u>Natural selection</u>**
    - Enhanced survival
    - Enhanced reproduction

Natural selection alters allele frequencies because they experience higher rates of transmission to offspring



How do you get changes in allele frequencies?

HWE is accurate in predicting allele frequency in the short term (generation-to-generation) rather than the long term (macroevolution)

Violations of the 5-assumptions will cause changes in allele frequency:

- Finite population sizes (genetic drift)
- Natural selection
- **Non-random mating**
    - Sexual-selection
    - Immunity diversification

How do you get changes in allele frequencies?

HWE is accurate in predicting allele frequency in the short term (generation-to-generation) rather than the long term (macroevolution)

Violations of the 5-assumptions will cause changes in allele frequency:

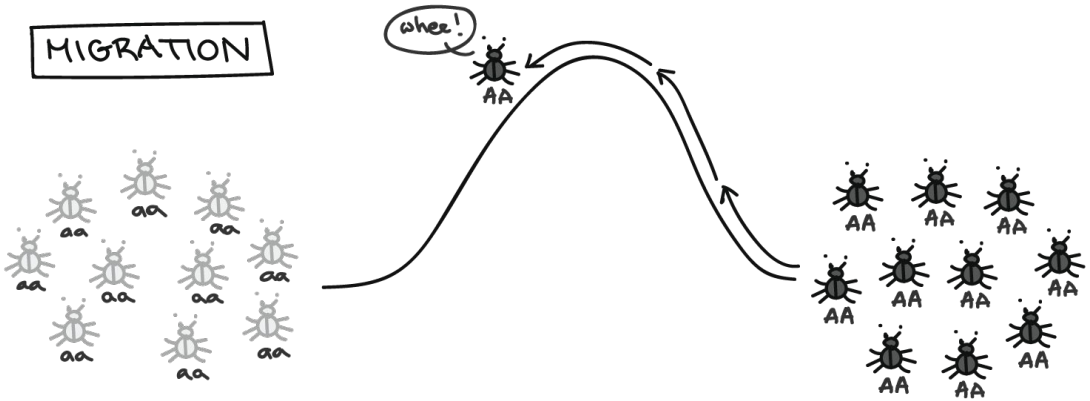
- Finite population sizes (genetic drift)
- Natural selection
- Non-random mating
- **Migration** (from other populations)

Migration introduces alleles to populations, altering steady-state allele frequencies, violating HWE

Different populations of a species may have different alleles at different frequencies

- Due to drift, selection, non-random mating, or mutation

Influx of these alleles violates HWE expectations → alleles not accounted for in p or q

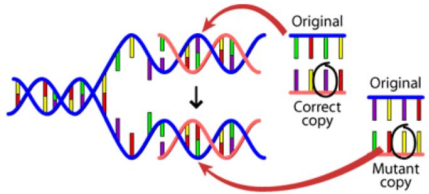


How do you get changes in allele frequencies?

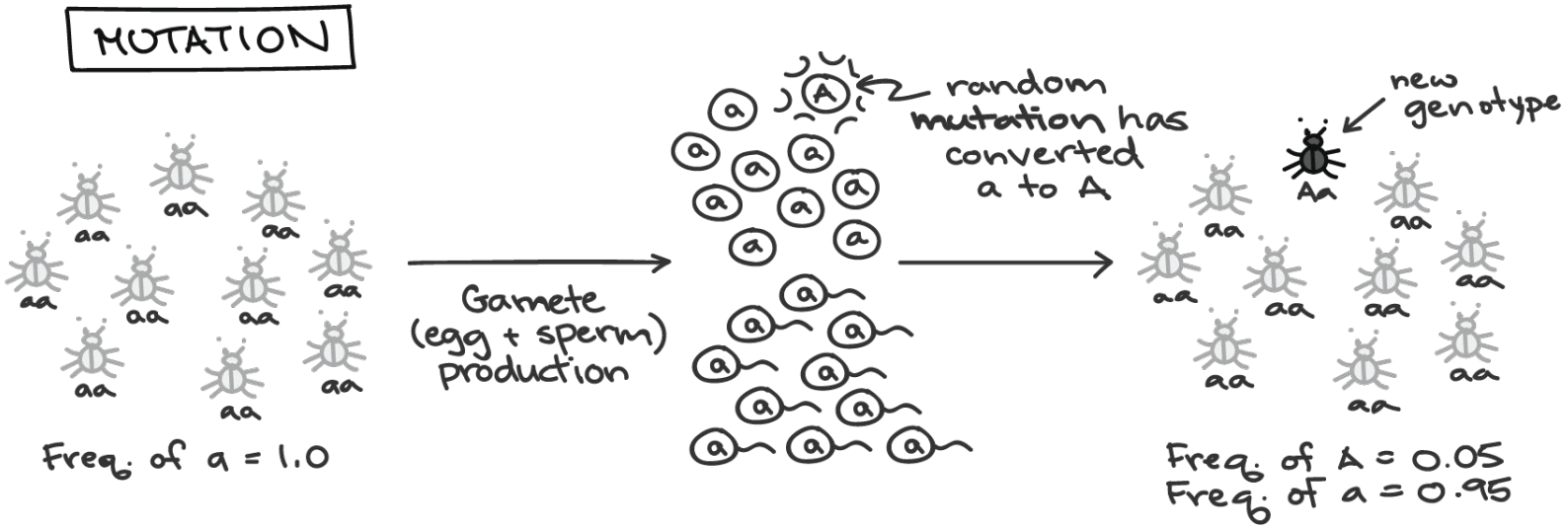
HWE is accurate in predicting allele frequency in the short term (generation-to-generation) rather than the long term (macroevolution)

Violations of the 5-assumptions will cause changes in allele frequency:

- Finite population sizes (genetic drift)
- Natural selection
- Non-random mating
- Migration (from other populations)
- **<u>Mutations</u>** - occur at a regular rate and are why HWE fails over the long-term



Mutations are the ultimate source of genetic novelty, but their creation violates HWE



No alleles at site

Biallelic site that can be tested for HWE in later generations